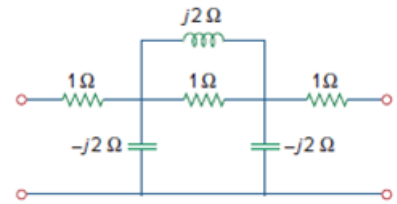


Problem

For the circuit shown in Fig. 19.129, use *PSpice* or *MultiSim* to obtain the t parameters. Assume $\omega = 1 \text{ rad/s}$.

Figure 19.129



Step-by-step solution

Step 1 of 2

(a) Since $a = \frac{V_2}{V_1} \Big|_{I_1=0}$ and $c = \frac{I_2}{V_1} \Big|_{I_1=0}$,

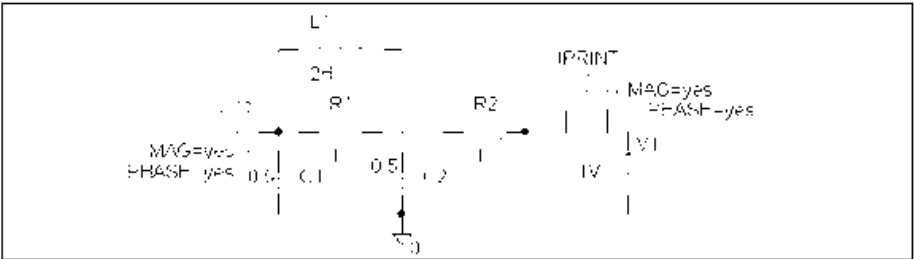
we open-circuit the input port and let $V_2 = 1 \text{ V}$. The schematic is shown below. In the AC Sweep box, set Total Pts = 1, Start Freq = 0.1592, and End Freq = 0.1592. After simulation, we obtain an output file which includes

FREQ	IM(V_PRINT2)	IP(V_PRINT2)
1.592 E-01	5.000 E-01	1.800 E+02
FREQ	VM(\$N_0001)	VP(\$N_0001)
1.592 E-01	5.664 E-04	8.997 E+01

From this,

$$a = \frac{1}{5.664 \times 10^{-4} \angle 89.97^\circ} = 1765 \angle -89.97^\circ$$

$$c = \frac{0.5 \angle 180^\circ}{5.664 \times 10^{-4} \angle 89.97^\circ} = -882.28 \angle -89.97^\circ$$



Step 2 of 2

(b) Similarly,

$$b = -\frac{V_2}{I_1}\bigg|_{V_1=0} \quad \text{and} \quad d = -\frac{I_2}{I_1}\bigg|_{V_1=0}$$

We short-circuit the input port and let $V_2 = 1$ V. The schematic is shown below. After simulation, we obtain an output file which includes

FREQ	IM(V_PRINT2)	IP(V_PRINT2)
1.592 E-01	5.000 E-01	1.800 E+02
FREQ	IM(V_PRINT3)	IP(V_PRINT3)
1.592 E-01	5.664 E-04	-9.010 E+01

From this, we get

$$b = -\frac{1}{5.664 \times 10^{-4} \angle -90.1^\circ} = -j1765$$

$$d = -\frac{0.5 \angle 180^\circ}{5.664 \times 10^{-4} \angle -90.1^\circ} = j888.28$$

Thus $[t] = \begin{bmatrix} -j1765 & -j1765 \\ j888.2 & j888.2 \end{bmatrix}$

